

Research Paper 12 Summary

Forest Fire Risk Assessment Using Machine Learning Methods

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Xu et al. (2020), *Remote Sensing*

1 Basic Information

- **Paper Title:** Forest Fire Risk Assessment Using Machine Learning Methods: Yunnan, China
- **Authors:** Rong Xu, Xiulin Lin, Shaoming Ke, and Chuanmin Hu (2020)
- **Core Topic:** Comparing different Machine Learning algorithms (Random Forest, SVM, Logistic Regression, Decision Trees) on multi-source geospatial data.

2 What is this Paper About? (In Simple Words)

This paper evaluates how well different AI/Machine Learning models perform when predicting and classifying fire events using multi-source satellite and environmental data.

- **The Data Fusion Approach:** Instead of using only raw satellite temperature, the authors combined three different types of data:
 1. **Topography / Geography:** Elevation, land slope, and distance to roads/settlements.
 2. **Vegetation Indices:** Normalized Difference Vegetation Index (NDVI) from satellites.
 3. **Weather / Climate:** Air temperature, humidity, and rainfall.
- **The Winning Algorithm: Random Forest** outperformed all other algorithms (achieving an AUC score > 0.88). It proved to be the most resilient model against noisy satellite data and missing sensor values.

3 Key Concept: Feature Importance Analysis

The paper uses Mean Decrease Impurity (Gini Importance) to rank which input features contribute most to accurate fire classification. This proved that combining satellite imagery with geographic context features produces much higher accuracy than using satellite data alone.

4 How This Helps Our Project (PS26162)

4.1 1. Scientific Justification for Using Tree-Based Ensembles

When evaluators ask: “*Why did you choose Random Forest or XGBoost instead of a basic Deep Neural Network or Logistic Regression?*”, this paper provides the peer-reviewed justification:

- Tabular geospatial data (combining coordinates, temperatures, distances, and land-use flags) is best handled by tree-based ensemble models.
- Tree models handle non-linear boundaries, missing data, and mixed data types (numbers + text categories) with high stability.

4.2 2. Blueprint for Multi-Modal Feature Engineering

Our project follows the exact feature engineering structure validated in this paper:

- **Thermal Features:** FIRMS Brightness Temperature, FRP, Day/Night flag.
- **Spatial Context Features:** Distance to nearest OSM industrial polygon, distance to road, built-up density.
- **Environmental Features:** Land cover class, NDVI vegetation score, seasonal month index.

4.3 3. Ranking Feature Importance in Our Pipeline

The paper provides the methodology for extracting and reporting feature importance scores, allowing our team to explain which inputs (e.g., proximity to industrial polygons vs. thermal intensity) contribute most to our model's decisions.